

1 Title

2 Detection and Neural Encoding of Whisker-generated Sounds in Mice

3 Authors

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7 Summary

8 The vibrissa system of mice and other rodents enables active sensing via whisker movements
9 and is traditionally considered a purely tactile system. Here, we ask whether whisking against
10 objects produces audible sounds and whether mice are capable of perceiving these sounds.
11 We found that whisking by head-fixed mice against objects produces audible sounds well within
12 their hearing range. We recorded neural activity in the auditory cortex of mice in which we had
13 abolished tactile sensation, and found that the firing rate of auditory neurons was strongly
14 modulated by whisking against objects. Furthermore, the object's identity could be reliably
15 decoded from the population neuronal activity, and closely matched the decoding patterns
16 derived from sounds that were recorded simultaneously with the neuronal activity, suggesting
17 that neuronal activity reflects acoustic information. Lastly, trained mice, in which tactile
18 sensation was abolished, were able to accurately identify objects solely based on the sounds
19 produced during whisking. Our results suggest that, beyond its traditional role as a tactile
20 sensory system, the vibrissa system of rodents engages both tactile and auditory modalities in
21 a multimodal manner during active exploration.

22 Acknowledgments

23 We thank E. Nelken, N. Ulanovsky, and Y. Livneh for valuable discussions, S. Devore for
24 valuable feedback and insightful comments on the manuscript, and R. Tumasus for the
25 histological preparations.

26 Funding

27 I.L. is the incumbent of the Norman and Helen Asher Professorial Chair at the Weizmann
28 Institute of Science.

29 Y.K. is the incumbent of the Marianne Manoville Beck Research Fellow Chair in Brain
30 Research.

31 This research was supported by the Pasteur-Weizmann Council from the Institut Pasteur and
32 the Weizmann Institute of Science (IL).

33 This work was supported by the Swiss Society Center for Research on Perception and Action at
34 the Weizmann Institute of Science.

35 This work was supported by the Israel Science Foundation grant ISF 1325/22 (IL).

36 This work was supported by the United States-Israel Binational Science Foundation (BSF)
37 2019251 (IL)

38 This work was supported by the Deutsche Forschungsgemeinschaft, SFB 1089 (IL)

39 **Author contributions**

40 I.L. and B.E. designed and conceptualized the experiments and hypothesis, B.E. and A.N.
41 performed the behavioral experiments, B.E. performed the electrophysiological experiments,
42 B.E. carried the analysis, I.L., B.E., Y.K., and A.N. wrote the manuscript.

43 **Declaration of interests**

44 The authors declare no competing interests.

45 **Declaration of generative AI and AI-assisted technologies in the writing process**

46 During the preparation of this work, the authors used ChatGPT 4 in order to improve the
47 language and readability of the manuscript. After using this tool/service, the authors reviewed
48 and edited the content as needed and take full responsibility for the publication's content.

49 **Supplemental information**

50 Document S1. Figures S1–S4

51 Video 1, Whisking states, related to Figures 1,3 and 4

52 Video 2, Sound-induced activity in two units, related to Figure 3

53 Video S1, Rhythmic sound generation example, related to Figure 3

54 **Data and materials availability**

55 All data, code, and materials used in this paper will be uploaded and made available online
56 upon publication.

57

58 Introduction

59 Sensory systems process unique information types, integrating diverse stimuli to enable
60 animals to perceive their environment multimodally, generating a coherent experience. This
61 integration is crucial for adapting behaviors and actions based on comprehensive
62 environmental assessments.

63 Acquiring certain types of sensory information requires specific actions, such as moving a finger
64 over a surface to sense its texture or inhaling to smell odors. Actions can generate sensory
65 information, as seen in echolocation¹ or electrolocation², or can be passively gathered from
66 environmental changes. For instance, many mammals actively probe their environment with
67 their whiskers to collect tactile information while passively using hearing to receive sound
68 waves.

69 Actions themselves can generate sounds through interactions with objects or other entities in
70 the environment. Sometimes actions are made to intentionally generate sound^{3–6}. For example,
71 visually impaired individuals use white canes to identify objects and avoid collisions⁷. As the
72 white cane moves, it produces distinct sounds depending on its contact surface, aiding in
73 surface identification⁸. Similarly, tapping on objects can provide auditory information about their
74 shape and material.^{9,10}

75 Somatosensation primarily relies on active movements to sense the environment and objects,
76 which can generate incidental sounds. While we know that incoherence between
77 somatosensory and auditory input can alter perception^{11–16}, we don't know if sounds produced
78 during active tactile sensation provide additional information about the objects. Investigating
79 actions that simultaneously generate auditory and tactile information can provide insights into
80 the brain's multimodal processing and integration mechanisms.

81 In rodents, the vibrissa system facilitates active exploration of the proximal environment, even
82 in complete darkness, through the movement of the whiskers^{17–19}. Whisking has typically been
83 considered a tactile sensation that involves the neural processing of tactile signals in the barrel
84 cortex. No studies have examined whether whisking produces acoustic signals. Like the
85 acoustic cues produced by using a white cane, we tested the hypothesis that the whisker
86 system enhances the sensation of the environment by producing audible cues.

87 Here, we show that the whiskers of mice, despite being very thin (base diameter of 40-80 μm
88 and 3-4 μm at the tip)^{20,21}, produce sound when contacting objects during active whisking.
89 Moreover, these sounds are reliably encoded in the auditory cortex (AC) and can be used
90 behaviorally by mice during object identification. Our study suggests a novel function for
91 auditory-tactile integration in natural rodent behavior.

92 Results

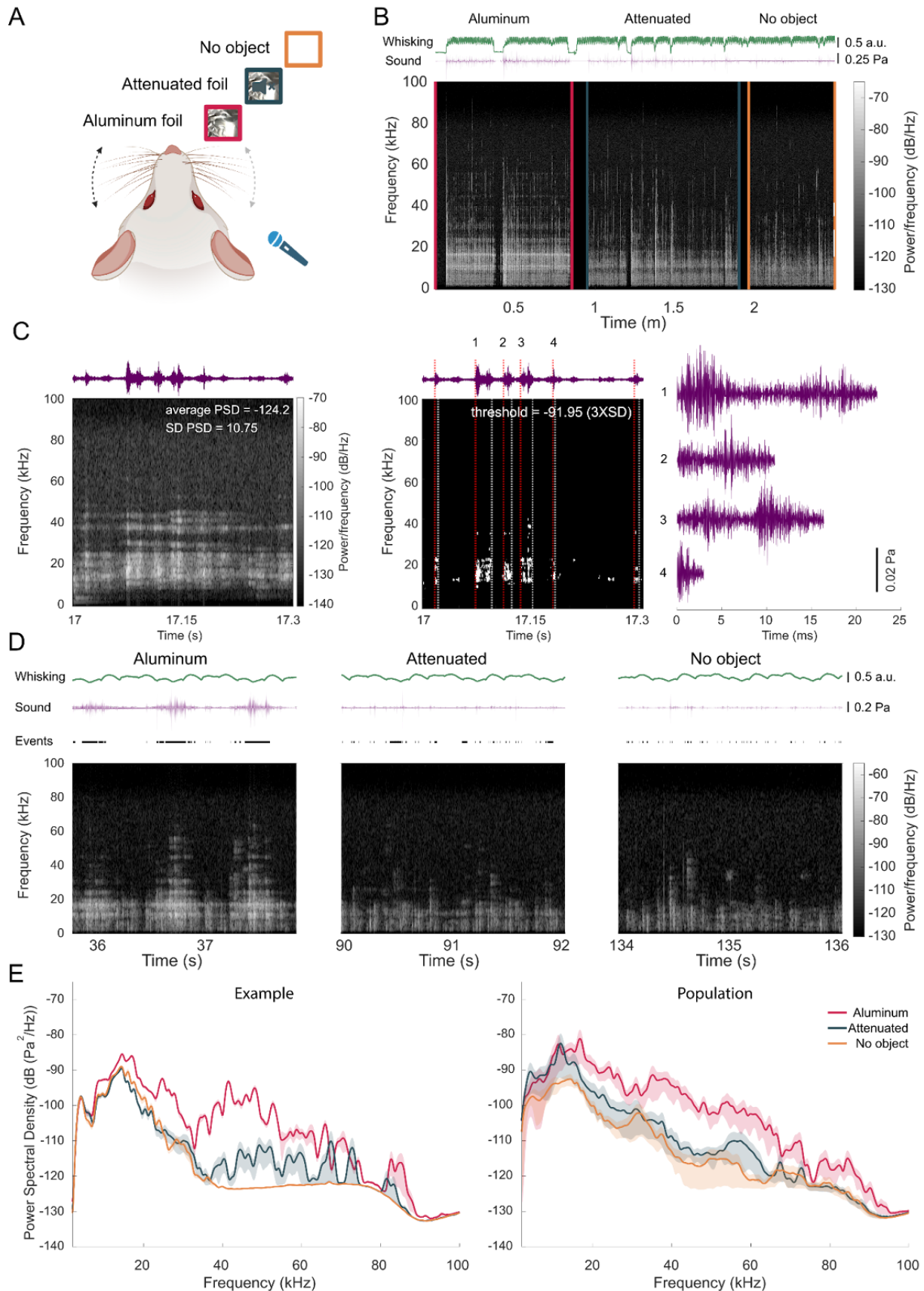
93 Whisking against objects generates sounds.

94 We first tested whether whisking against different objects produces distinct acoustic signals with
95 properties that could be behaviorally relevant for mice. To this aim, we made recordings using
96 an ultrasonic microphone while head-fixed mice whisked freely against different surfaces,
97 including aluminum foil, sound-attenuated aluminum foil (hereafter referred to as 'attenuated
98 foil,' see Methods), as well as against no-object (Figure 1A). The ultrasonic microphone was
99 positioned 2 cm from the whiskers, roughly the same distance as the whiskers to the mouse's
100 ear. We used FVB/NHanHsd mice due to their lower and less variable sound threshold

101 compared to C57BL/6 mice²²⁻²⁴, making them suitable for exploring complex auditory
102 responses. We extracted whisker movement energy from high-speed (400 fps) and
103 high-contrast videos and identified whisking periods through video analysis (see Methods).
104 We extracted sound events during whisking (Figure 1B, C) based on the spectral content
105 (Figure 1C). We analyzed the sounds separately for each object and the no-object condition.
106 We calculated the mean spectrogram for all events from each mouse (Figure 1E left panel) and
107 averaged them across all animals (n = 5, Figure 1E right panel). Although there were individual
108 differences in the spectrum of sound events between mice, there are general trends in the
109 spectral content of sound events across mice. Aluminum foil events (pink) had higher energy
110 across the spectrum for frequencies above 10 kHz compared to the no object and above 15
111 kHz compared to the attenuated foil. Comparing the attenuated foil to the no object, we see a
112 peak at ~10 kHz and in the 50-60 kHz range. We further compared the aluminum and
113 attenuated foil whisking sounds to sounds generated by whisking against dry Bougainvillea
114 leaves (Supplementary Figure 1A-B). Using the leaf, we showed that natural objects can
115 generate whisking-related sounds at audible levels and within vocalization amplitudes²²⁻²⁸ as
116 well and that the sounds differ between objects.
117 To rule out the possibility that differences in the amplitude of generated sounds for the different
118 objects might arise from increased vigor in whisking, we compared motion energy derived from
119 the whisking videos across different objects (Supplementary Figure 1C-D). Differences in
120 whisking across animals (Supplementary Figure 1D) were minimal, and even in cases where
121 we observed significant differences in whisking between different objects, these did not
122 correlate with significant differences in the overall sound intensity (Supplementary Figure 1C). A
123 possible explanation for the observed differences in whisking energy could be that some of the
124 objects (i.e., aluminum and attenuated foils) are heavier and more rigid than others (i.e., the
125 leaf), restricting some of the movement of the whiskers. Overall, our findings reveal that
126 whisking against different materials produces unique auditory signatures, with amplitudes and
127 frequencies in the audible range of mice (Supplementary Figure 1A)²²⁻²⁸.

128

129



130

131 Figure 1 – Characterization of the sounds during whisking against objects.

132 (A) Sound recording configuration from head-fixed mice whisking against aluminum, attenuated, and no
 133 object. The sonic and ultrasonic microphone was positioned ~ 2 cm from the snout. In all panels: pink,
 134 aluminum; gray, attenuated; yellow, no object; green line, whisking; purple line, Objects were switched
 135 using a motor.

136 (B) An example of continued sound recording, showing the whisking trace, the recorded sound
137 amplitude, and the power spectral density (PSD) of the recorded sound across time for the 3 conditions
138 which their starting and ending times marked by the colored vertical lines.

139 (C) Left: Raw PSD of recorded whisking sounds against aluminum. purple, recorded sound. Middle:
140 Binary mask of the left panel segment, created by thresholding the PSD. white, above threshold; black,
141 below threshold. Events were identified based on minimal length and frequency bands above the
142 threshold (see Methods). red dashed line, start of event; white dashed line, end of event. Right: Zoomed
143 waveform of the four numbered events in the middle panel.

144 (D) Zoom in on the recording in the three conditions. Left: Aluminum foil. Middle: Attenuated. Right: No
145 object. Sound events were identified and are marked by black lines.

146 (E) Mean PSD of identified sound events between the objects during the whisking periods $n = 5$.

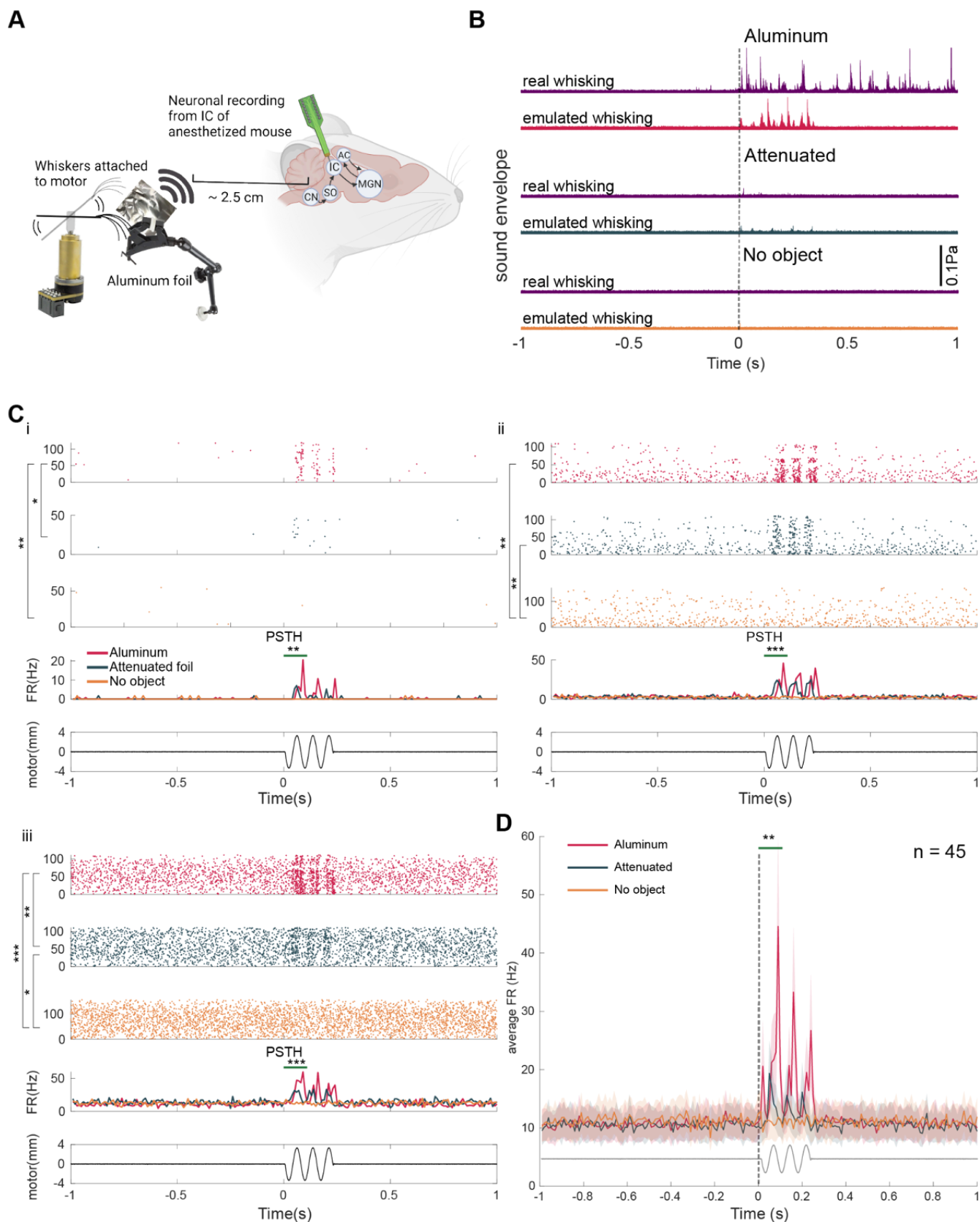
147 Data are represented as mean $\pm$ SEM.

148

149 **Neuronal Responses in the Inferior Colliculus (IC) Triggered by Emulated Whisking**

150 To investigate if the sounds that are produced when mice whisk against objects can elicit
151 neuronal activity in the auditory pathway, we attached 6 plunked whiskers to a motor and
152 moved them against objects, emulating natural whisking motions (11 Hz, 30°). We recorded
153 extracellular responses from the IC of anesthetized C57BL/6 mice ($n = 2$) with their left ear
154 positioned 2 cm from the emulating whisking apparatus with the aim of producing sounds with
155 similar intensity to those produced during natural whisking (Figure 2A). Therefore, before
156 recording the neuronal activity, we tuned the emulated whisking (EW) to produce sound
157 intensities similar to the sound that is produced during natural active whisking (AW). Sound
158 amplitudes were comparable between active whisking and emulated whisking at the speed we
159 used (Figure 2B).

160 While emulating whisking, we recorded extracellular IC neuronal activity from anesthetized
161 C57BL/6 mice under three conditions: aluminum foil, attenuated foil, and no object. We isolated
162 the spiking activity of 45 units from $n=2$ mice. The response of the two example units (Fig. 2Ci
163 and ii) displayed significant differences in firing rate (FR) between the three conditions, with the
164 highest FR when whisking was emulated against the aluminum foil, a lower FR for the
165 attenuated foil and the lowest for the no-object. These differences were also found at the
166 population level (Figure 2D), we observed significant differences in the Δ FR measured 110 ms
167 after motor onset compared to baseline measured before the motor onset (Friedman test, $p =$
168 $5.27e-8$). Aluminum showed a higher Δ FR ($8.95 \text{ Hz} \pm 20.29$) compared to attenuated ($2.18 \text{ Hz} \pm$
169 5.22 , $p = 0.011$) and no object ($0 \text{ Hz} \pm 1.311$ $p = 2.11e-8$). Attenuated also showed higher Δ FR
170 compared to no object ($p = 0.011$). These findings confirm that whisking sounds can elicit
171 neuronal activity in the IC, an early auditory processing stage. Because we positioned the
172 objects at a similar distance of their own whiskers to their ears, the neuronal responses to
173 emulated whisking in anesthetized mice suggest that during natural behavior, sounds elicited
174 when mice whisk against objects are represented in the auditory system. This provides a basis
175 for further investigation into their representation in the AC of awake-behaving mice.



176

177 **Figure 2** – Auditory response of neurons in the Inferior Colliculus of anesthetized mice to emulated
178 whisking against objects when positioned near the ears of the nearby mice.

179 In all panels: pink, aluminum; gray, attenuated; yellow, no object.

180 (A) A motor maneuvered the attached plucked whiskers against various objects placed 2 cm from the
181 mouse ears, akin to the natural distance of the whiskers from the ear. Neuronal response was recorded
182 using a silicon probe in the IC.

183 (B) Comparison of the sound amplitude envelope between the emulated and active whisking. The top
184 traces (purple) depict recordings from an awake head-fixed mouse whisking against objects. The bottom
185 traces show the emulated whisking recordings: top - aluminum (pink), middle - attenuated (gray), and
186 bottom - no object (yellow). Time 0 indicates the start of motor movement (bottom traces) or the initiation
187 of whisking (top traces). For whisking against aluminum foil (top panel), AW had a mean amplitude of
188 0.063 Pa (± 0.074 Pa SD), with a peak of 0.109 Pa. EW (pink) showed a mean amplitude of 0.058 Pa
189 (± 0.005 Pa SD), with a peak of 0.079 Pa. For whisking against attenuated foil (middle panel), AW had a
190 mean amplitude of 0.003 Pa (± 0.001 Pa SD), with a peak of 0.023 Pa; EW (gray) had a mean amplitude
191 of 0.003 Pa (± 0.002 Pa SD), with a peak of 0.021 Pa.

192 (C) Units 19 (i), 11 (ii), and 9 (iii) responded to emulated whisking upon contact with aluminum and
193 attenuated foils but showed no response to emulated whisking in the absence of an object. The top
194 three panels in (i) and (ii) are raster plots with time 0 corresponding to the start of the motor movement.
195 The motor moves back and forth three times. Beneath the raster plots, the peristimulus time histogram
196 (PSTH) is presented, and spiking activity was binned into 10-ms bins. The final panel illustrates the
197 motor's movement.

198 (D) The mean response of all isolated units following the onset of the motor movement. The shaded
199 regions denote $\pm$ SEM. * $p < 0.05$, ** $p < 5e-5$, *** $p < 5e-10$.

200

201 **Auditory cortex neurons exhibit diverse responses to sounds generated when mice** 202 **whisk against objects.**

203 Next, we examined if AC neurons encode the sound generated when mice whisk against
204 different objects. Therefore, we recorded neuronal activity from AC of head-fixed mice ($n = 5$)
205 running on a rotating treadmill, allowing their whiskers to make contact with aluminum foil,
206 attenuated foil, and no object (Figure 3A). To diminish the potential confounding involvement of
207 the somatosensory system in the activation of neurons in the AC upon whisking against objects,
208 we severed the mouse's infraorbital nerve (ION), which conveys tactile sensations from the
209 whisker pad and cheek (Supplementary Figure 3). To discern if changes in firing rates were due
210 to motor commands, we compared neuronal activity elicited by aluminum foil to that of
211 attenuated foil, where the major difference between these two objects is the sound that is
212 produced upon whisking. The motor switched between objects (aluminum, attenuated, and no
213 object) at fixed intervals, moving them toward the whiskers, and whisking behavior was
214 monitored with a high-speed camera (Figure 3A). We identified two behavioral states (see
215 Methods, Video 1) in the experiment (Supplementary Figure 2): non-whisking state (NWS) and
216 active whisking state (AWS). We observed changes in neuronal firing rate between the two
217 whisking states, and we also observed additional differences in FR between the objects
218 presented to the whiskers in the AWS, with these changes varying across the population
219 (Figure 3B). In the example (Figure 3B, Video 2), unit 122 (orange trace) showed moderate yet
220 significant elevation in FR between the NWS (green shade) and the AWS for the no-object
221 (yellow shade) conditions (1.67 Hz versus 3.75 Hz). Whisking against aluminum foil (pink
222 shade) drastically increased the FR of the same unit (16.42 Hz). However, when the mouse
223 was whisking against attenuated foil (gray shade), the FR significantly dropped (2.99 Hz,
224 slightly below the FR during whisking against no object), strongly suggesting that sounds

created by whisking against aluminum foil, activated unit 122. Conversely, unit 144 (blue trace) was activated by whisking but was inhibited by the sound generated by the aluminum sound, whereas its FR during NWS (4.86 Hz) dramatically increased during AWS without an object (10.78 Hz) or when the mouse was whisking against attenuated foil (13.36Hz), its FR was strongly decreased when the mouse was whisking against aluminum (3.60 Hz). Note that the aluminum sounds suppressed unit 144 activity even below the NWS baseline.

To better understand the temporal relationship between sound events and neuronal firing rates (FR) in the auditory cortex (AC), we identified specific sound events. We constructed peristimulus time histograms (PSTHs) around their onset (Figure 3C). Unit 28 exhibited a sharp increase in FR following aluminum sound events, with a smaller response to attenuated sound events. A comparison of average activity around these sound events revealed significant differences across conditions. Specifically, FR around aluminum events was higher compared to both attenuated and no-object sound events. In contrast, Unit 30 displayed a different pattern. While it also showed an activity peak following aluminum sound events, the overall FR around aluminum events was lower compared to attenuated and no-object events.

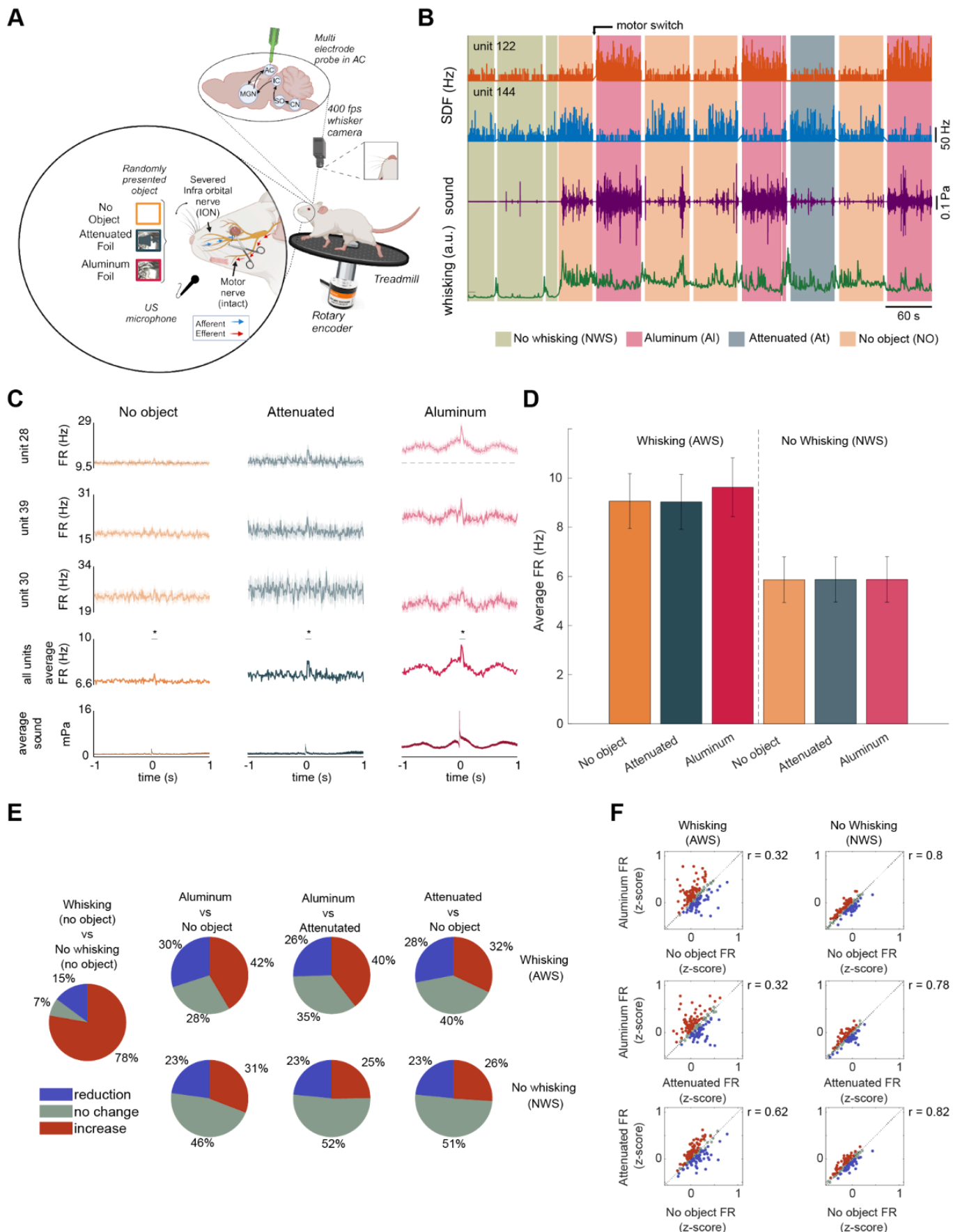
The average response ($n = 16$ units for which sufficient sound events were present for statistical analysis) locked to the sound events was significantly higher than the baseline (measured from a time window before the event onset) for aluminum sound events ($\Delta\text{FR } 1.25 \text{ Hz} \pm 1.46$, mean $\pm$ SD; Wilcoxon signed-rank test, $p = 8.79\text{e-}4$) and attenuated foil sound events ($0.45 \text{ Hz} \pm 0.49$; $p = 2.41\text{e-}4$) and the no object sound events ($24.03 \text{ Hz} \pm 6.92$; $p = 3.9\text{e-}3$).

The average FR around aluminum sound events shows a sinusoidal pattern with a frequency of $\sim 1.5 \text{ Hz}$. This pattern is also apparent in the averaged envelope of all sound events in the aluminum foil, suggesting that the neuronal activity represents the sound amplitude. From manual observation of the whisking videos (Supplementary Video S1), there is a periodic $\sim 1.5 \text{ Hz}$ movement cycle of the entire whisker pad, nose, and ears. Next, we wanted to understand how the mean FR of the entire recorded population of neurons represents the different whisking states (AWS and NWS) and the different objects.

Comparing all units' average FR (Figure 3D), the only significant difference ($p = 9.56\text{e-}7$, Wilcoxon rank-sum test) was between AWS FR ($9.23 \pm 8.1 \text{ Hz}$, mean $\pm$ SD) and NWS ($5.92 \pm 5.97 \text{ Hz}$). While differences between conditions within each whisking state were not statistically significant (Friedman test, $p = 0.1374$), the average firing rate in AWS was slightly higher for aluminum ($9.62 \pm 5.97 \text{ Hz}$) compared to attenuated ($9.02 \pm 8.01 \text{ Hz}$) and no object ($9.06 \pm 8.05 \text{ Hz}$). However, the examples shown in Figure 3C suggest that individual units exhibited specific modulation. Thus, we explored the activity differences for each unit (Permutation test, 10,000 permutations, $p < 0.05$ after Tukey-Kramer HSD correction, Figure 3E-F). Our analysis revealed that under the no-object condition, whisking prominently elevated FR. Most units showed significant changes in FR, with most showing elevation during whisking against no object compared to NWS (Figure 3E, left). Comparing different object conditions (aluminum and attenuated) to no object, more units showed significant changes in FR between aluminum and no object than between attenuated and no object in the AWS (Figure 3E, right top). Moreover, a similar trend was also found in the NWS, but for this state, a larger fraction of units was not modulated by the object conditions (Figure 3E, right bottom). Overall, during AWS more units showed increased FR with aluminum foil compared to attenuated foil.

Next, we examined the modulation of each unit across states and conditions. In AWS, we observed lower correlations (Figure 3F) between conditions (Pearson correlation coefficients: 0.32, 0.31, 0.62; aluminum-no object, aluminum-attenuated, attenuated-no object) than in NWS

272 for all object comparisons (0.79, 0.78, 0.82). In both whisking states, FRs for attenuated and no
273 object were more correlated with each other (Figure 3F, bottom panels) than with aluminum.
274 The fact that FR of the units in NWS scatter near the identity diagonal line implies minimal
275 changes in firing rates between object conditions. In contrast, during whisking (AWS), the firing
276 rates of many units were strongly affected by the objects' nature, either increased or reduced,
277 which overall reduced the correlations across object conditions.
278 Collectively, the diverse modulation of neuronal firing rates in the AC by different objects during
279 whisking - including both increases and decreases in activity - raises an intriguing question:
280 Could these neurons collectively encode the specific types of objects presented to the mice?



281

282 **Figure 3** – Neurons in AC exhibit diverse responses to whisking states and objects.

283 (A) Awake mice, head-fixed on a rotary treadmill, were exposed to various objects positioned near their
 284 whiskers, including aluminum foil (pink), attenuated foil (gray), and no object (yellow). Sounds were

recorded via an ultrasonic microphone positioned ~2 cm from the ears. Neuronal activity was recorded from the AC using a silicon probe, while whisking behavior was monitored with a high-speed camera.

(B) Activity patterns of two AC units during exposure to the three objects (represented by shaded areas) in a single session. Initially, without whisking (green trace), Unit 122 (orange trace, SDF) displayed an increased FR when whisking against aluminum, corresponding with a surge in sound amplitude (purple trace). Conversely, Unit 144 (orange trace, SDF) showed a low baseline activity level. The FR increased during whisking and decreased below baseline upon presentation of aluminum (pink shade). —see Video 2.

(C) Example of three other units showing their mean response to identified whisking sound events for the three conditions during whisking: Left: no object; middle: attenuated; right: aluminum. **Unit 28:** peak FR for aluminum was $27.62 \text{ Hz} \pm 0.88$ (30 ms after event onset), and for attenuated foil was $17.38 \text{ Hz} \pm 1.16$ (40 ms). The average FR (± 1 second around the sound events) was $18.72 \text{ Hz} \pm 3.97$ for aluminum, $12.55 \text{ Hz} \pm 3.66$ for attenuated foil, and $5.7 \text{ Hz} \pm 18.9$ for no object (Kruskal–Wallis test, $p < 1e-32$). **Unit 39:** peak FR for aluminum was $29.87 \text{ Hz} \pm 1.04$ (30 ms), and for attenuated foil, it was $23.18 \text{ Hz} \pm 1.39$ (30 ms). The average FR (± 1 second around the sound events) was $23.23 \text{ Hz} \pm 5.32$ for aluminum, $17.98 \text{ Hz} \pm 5.32$ for attenuated foil, and $17.01 \text{ Hz} \pm 6.33$ for no object (Kruskal–Wallis test, $p < 1e-32$). **Unit 30:** peak FR for aluminum was $26.07 \text{ Hz} \pm 1.03$ (40 ms), and for attenuated foil, it was $32.66 \text{ Hz} \pm 1.94$ (360 ms). The average FR (± 1 second around the sound events) was $21.88 \text{ Hz} \pm 8.76$ for aluminum, $26.33 \text{ Hz} \pm 5.46$ for attenuated foil, and $24.03 \text{ Hz} \pm 6.92$ for no object (Kruskal–Wallis test, $p < 1e-32$).

Dashed lines depict the mean FR for the no-object condition. The bottom panel shows the average ($n = 16$) sound envelope of whisking sound events. * $p < 0.05$, ** $p < 5e-5$, *** $p < 5e-10$.

(D) Average FR for all objects in both whisking states. NWS - $n = 162$ units. AWS $n = 200$ units. Error bars stand for the 95% CI.

(E) Pie plots presenting the percentage of units that significantly increased (red) or decreased (blue) FR between the conditions. Left: comparison between AWS when whisking against no object to NWS with no object. Top: comparison within the AWS conditions between aluminum and no object (left), aluminum and attenuated (middle), and attenuated and no object (right). Bottom: similar comparison within the NWS.

(F) Normalized FRs (z-scored) of all recorded units in AWS (left panels) and NWS (right panel). Top: FR in the aluminum condition (y-axis) plotted against FR for the no object condition (x-axis). Middle: FR in the aluminum (y-axis) plotted against FR in the attenuated condition (x-axis). Bottom: FR in the attenuated condition (y-axis) plotted against FR in the no object condition (x-axis). The units are colored to indicate a significant increase (red) in FR when comparing the y-axis condition to the x-axis condition, a significant decrease (blue), and units that did not show a significant change (gray) in FR. Permutation test, 10,000 permutations, $p < 0.05$ after Tukey-Kramer HSD correction.

The neuronal activity and recorded sound predict whisking states and object conditions.

To explore the possibility that neurons in AC collectively encode whisking states and object identity, we applied a non-linear classification model. More specifically, we sought to determine

327 if we could train a model to decode these different conditions from the neuronal population
 328 activity of AC neurons.
 329 We employed random forest classifiers based on neuronal activity and analyzed the interplay
 330 between object types, whisking states, and neuronal activity.
 331 We trained the classifier on 80% of the data, while the remaining 20% were used for testing.
 332 Data was binned into an 800 ms bin sliding window with 400 ms steps.
 333 We labeled each time bin based on the experimental condition. Initially, we pooled all NWS data
 334 bins across conditions for the main model, treating them as a single label based on
 335 observations that different NWS conditions resemble each other more closely than
 336 corresponding AWS conditions (Figure 3D-F). One mouse (Mouse 4; Supplementary Figure 2)
 337 whisked continuously throughout the duration of the experiment hence there were no
 338 non-whisking periods, so we trained the model exclusively on object identity during AWS.
 339 We first examined how neuronal population activity encodes the identity of objects and the
 340 whisking state. Predictions from all sessions were pooled and compared to the given (true)
 341 labels (Figure 4B, upper left panel). The FR classifiers accurately predicted all classes
 342 (Aluminum, attenuated, no object, and no whisking). The most accurate predictions were for the
 343 NWS (94.1% True Positive (TP) rate) and the aluminum condition (86.8% TP rate). In
 344 comparison, predictions for the attenuated (75.9% TP rate) and no object (73.2% TP rate)
 345 conditions were less precise. Misclassified bins of attenuated condition were most frequently
 346 labeled as no object (16.1%), while misclassified bins of the no object condition were most often
 347 labeled as attenuated (14.8%). We also compared the performance with classifiers trained on
 348 shuffled labels to assess the chance level of prediction, which centers around the expected
 349 chance level of 25% (Figure 4B, top panels).
 350 The success of the FR-based classifiers raises questions about the nature of the data. Does the
 351 neuronal population encode the sound space of whisking against objects, or does it represent
 352 other information? For example, neuronal activity may represent an internal state of whisking
 353 generated by motor efferent-copy or neuromodulatory effects. Moreover, it is possible that other
 354 sensory cues were also involved in representing the objects. To address this issue, another set
 355 of classifiers was trained on the same data structure (time bins and labels), using features
 356 extracted from the sound recording as the predictors in the model. We assumed that the
 357 sound-based classifier would perform well and even better than the FR-based classifier. If the
 358 FR-based classifier performance tracks that of the sound-based classifier across conditions, it
 359 will strongly support the hypothesis that neuronal firing is affected mainly by sound.
 360 To construct the sound-based model, we extracted features from the sound recordings in both
 361 the spectral and amplitude domains (Figure 4A)^{29–31}. We took advantage of the ability to extract
 362 the same sound features from all sessions across all mice, creating a single classifier using all
 363 the available data. This model generalizes over both animals and objects as a slightly different
 364 variant of the objects was used for different animals. The sound-based classifier performed
 365 exceptionally well predicting all classes (Figure 4B, lower left panel), outperforming the FR
 366 classifiers in all categories except the attenuated (70.5% TP rate). The general trend seen in
 367 the FR classifiers is kept in the sound-based classifier. Similar to the FR classifier, the
 368 sound-based classifier was most accurate in predicting no whisking (96.5% TP rate), followed
 369 by aluminum (93.6% TP rate), and was less accurate in predicting attenuated and no object
 370 conditions (78.3% TP rate). Also, when looking at misclassifications of the attenuated and no
 371 object, we see that misclassified attenuated bins were most often labeled as no object (23.2%)
 372 and misclassified no object bins were most often labeled as attenuated (15.4%), similar to the
 373 FR-based classifiers.

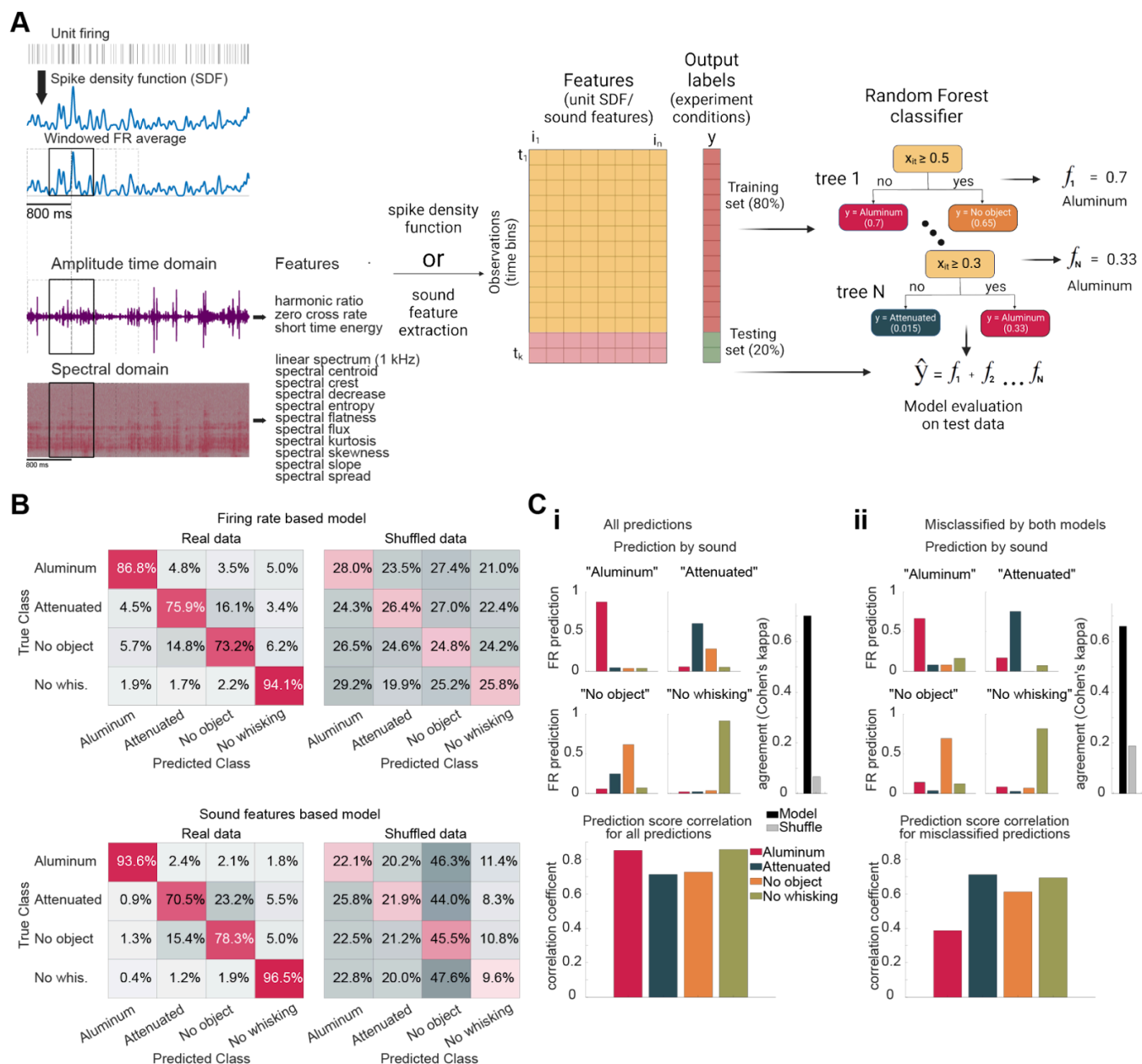
Next, we aimed to explore if the small differences in FR between objects that were observed in the NWS (Figure 3E) can also be decoded and classified based on neuronal activity and sound features. We examined the whisking videos (Supplemental video 1) and the whisking analysis and observed small twitches of the whisker pad under NWS (Figure 3B and Supplemental Figure 2A). Whereas in the above analysis, we pooled all the objects at the NWS, we decided to examine the possibility that such small whisker movements generated enough sound to drive distinct firing in AC for each object. Both the FR-based and sound feature-based classifiers were able to classify the objects above chance level (33.3%) correctly and better than shuffled data (Supplementary Figure 5).

When directly comparing the predictions from the FR-based and sound-based classifiers for the AWS and pooled no-whisking (Figure 4B), the two classifiers have a high agreement (0.70 Cohen's kappa, Figure 4Ci) on the labels they assign. The predictions made by the FR-based classifier were examined when the sound-based classifier predicted a specific label. For all sound-based aluminum predictions, 83% of the FR-based model's predictions were also for aluminum. The prediction scores (Figure 4Ci, top left) for the aluminum label from both models were highly correlated (0.87 r , $p < 1e-32$, Pearson correlation coefficient). When the sound-based model labeled an observation as no-whisking (Figure 4Ci, bottom right), 91% of predictions from the FR-based model also said no whisking. The prediction scores correlation was also high (0.88 r , $p < 1e-32$). Per the results seen when comparing the predictions to the true labels in both classifiers (Figure 4Ci), we see more uncertainty for the attenuated (Figure 4Ci, top right) and no object (bottom left). When the sound-based model labels an observation as attenuated, 60% of the FR-based model's predictions were attenuated, and 28% were no object. When the sound-based model labels an observation as no object, 62% of the FR-based model's predictions were no object, and 25% were attenuated. Despite being lower than the other labels, the prediction score correlations for both attenuated and no object remain high (0.73 r and 0.74 r , respectively, $p < 1e-32$). The FR-based and sound-based classifiers showed high agreement in their predictions, particularly for aluminum and no-whisking labels, with strong correlations in their prediction scores. Considering the generally congruent results and to further strengthen the claim that neuronal activity is driven by sound, we investigated scenarios where both models make mistakes to determine if they tend to make the same mistakes. We took a subset of the predictions where both models failed to predict the true label. When comparing misclassified predictions, agreement remained high (0.66 Cohen's kappa, Figure 4Cii). Moreover, for the no object, attenuated foil, and no whisking, the FR-based classifier tends to make the same mistake as the sound-based classifier, with a similar confidence level as shown by the prediction score correlations.

Together, these results suggest that the models generally capture similar trends in the data. The models were particularly reliable and accurate in predicting aluminum and no whisking, as indicated by the consistently high prediction scores; conversely, they tend to confuse attenuated and no object more frequently, leading to lower prediction accuracy for these classes. Given that the attenuated object produces lower sound intensity during whisking (Figure 1), it is likely more similar to the no-object condition with respect to the properties of the sound signal. This analysis emphasizes the models' strengths in specific scenarios and highlights areas for potential refinement in predicting complex auditory-induced neuronal responses.

In conclusion, our results confirm that the AC encodes distinct object identities through whisking-induced sounds and differentiates between various whisking activity states. These results support our main hypothesis that AC neurons can encode specific object types based on the auditory cues generated by whisking. The effectiveness of both the FR-based and

421 sound-based classifiers in accurately predicting object identities and whisking states highlights
 422 the AC's robust sensory processing capabilities when sound is produced upon whisking against
 423 different objects. .
 424



425
 426 **Figure 4** - The neuronal activity and recorded sound encode whisking states and object identity for all
 427 conditions

428 (A) We extracted the spike density function or the sound feature, binning them into 800-ms length
 429 windows. Bins were labeled based on the object identity in the AWS the 'No whisking' was given to all
 430 NWS bins regardless of the present object. We split the data into a training and testing set (80%-20%).
 431 A random forest model was trained on the training set, and its performance was evaluated on the testing
 432 data.

433 (B) A confusion matrix showing the pooled prediction accuracy percentage of the FR-based (top)
 434 classifiers from all sessions ($n = 7$) and the pooled sound-based classifier (bottom). The pink diagonal
 435 shows the correct classification; a darker shade indicates a more accurate prediction. Grey off-diagonal

squares indicate misclassifications, with a lighter shade representing fewer mistakes. Left: actual labels for each bin; right: shuffled labels.

(C) Comparison of all the predictions made by the sound-based and FR-based models for the same data bins. Top: ratio of observations where the FR-based model predicts the same label as the sound-based model; the right part shows Cohen's kappa score for agreement: black: true labels, gray: shuffled labels. Bottom: The correlation coefficient for each class is based on the prediction score from all the observations, regardless of whether they were ultimately categorized as that class or not. (i) All predictions, (ii) subset of the observations where both the sound-based and the FR-based models failed to classify the correct class.

Mice detect objects using the sound that is generated when whisking against them.

Having established that whisking against objects generates a sound that is encoded in auditory cortical population activity, we next sought to determine if mice can use these sounds to guide their behavior. To test this, we used mice with severed ION on one side to eliminate vibrissa and facial tactile signals ($n=5$) and asked if we could train them on a go/no-go task. Mice were conditioned to lick a waterspout when presented with aluminum foil and to refrain from licking in the absence of any object (Figure 5A). Mice were trained in complete darkness to remove visual cues. Moreover, we frequently changed the objects and thoroughly cleaned instruments between sessions to prevent olfactory cues.

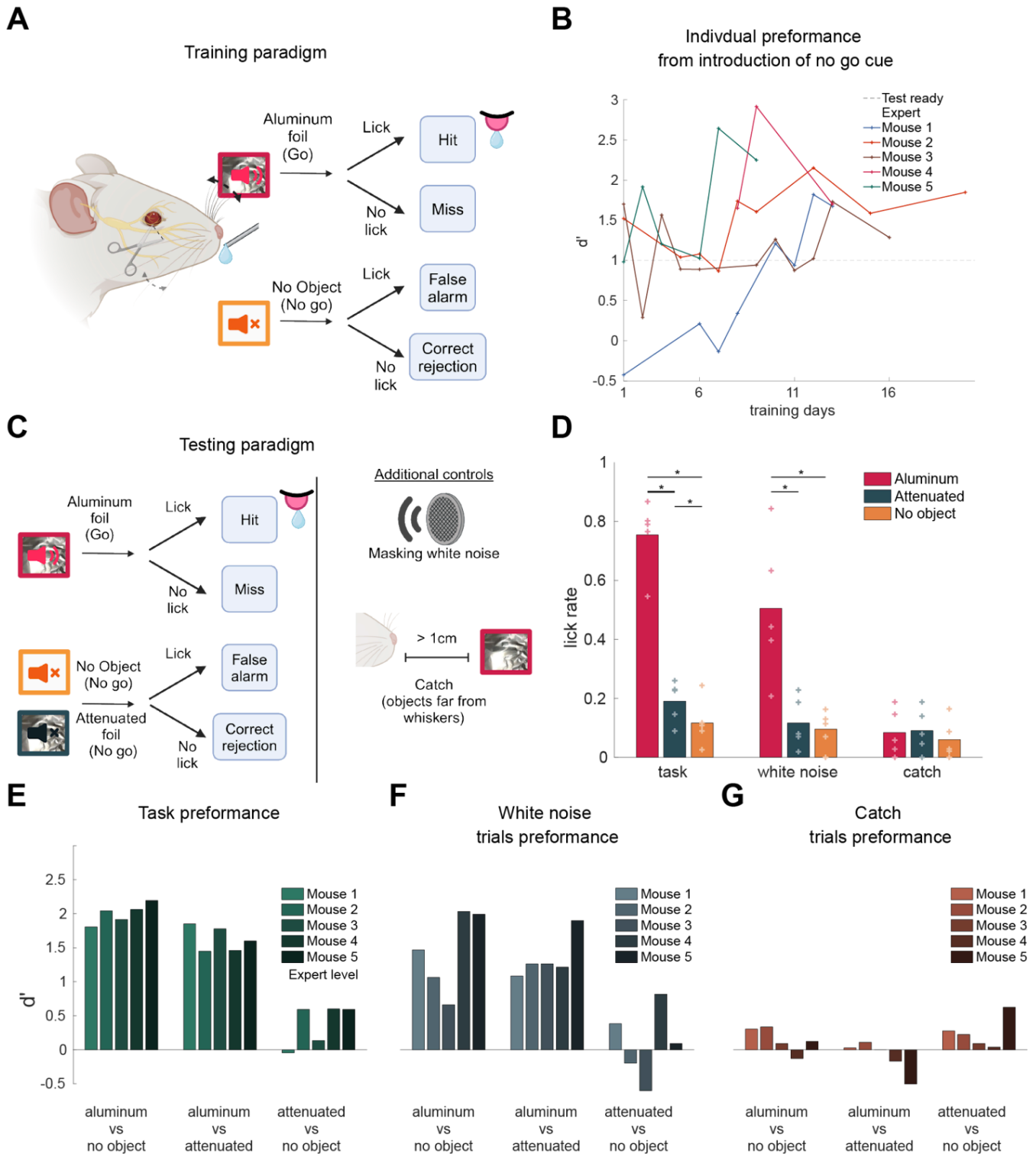
Despite lacking any tactile sensation through their whiskers, mice rapidly learned the task (Figure 5). Initially, mice were trained only on the 'go' cue and learned to lick at the correct time window (2 seconds) when the motor with the aluminum foil reached the whisker pad, and no other objects were introduced. A long inter-trial interval (>10 sec) was introduced to avoid constant licking between trials. Then, a 'no go' (no-object) cue was also included (Figure 5B, 'day 1'), and it took the mice between 6-12 days to consistently achieve the desired level of performance ($d' = \sim 1.5$; Figure 5B). This suggests that mice quickly learned the task and could detect the 'go' cue. Note that the initial performance for day 1 was already high due to the training procedure, which started with only the 'go' trials and gradually progressed to the introduction of 'no-go' trials (see Methods). Four out of the five mice progressed to introducing the 'no go' trials in 2-3 days. For mouse 3, it took over two weeks to lick consistently for the 'go' trials.

Our results show that mice were able to identify the aluminum foil without tactile sensation, which suggests that they learned the task based on the whisking sounds. In the testing sessions, we also introduced the attenuated foil object, and to verify that the mice did not use other sensory cues in subsets of sessions, we combined the three object conditions (i.e., aluminum, attenuated foil, no object) with several controls (Figure 5C). The control trials (30% of the trials of a given session for which one type of control was used) included: 1) a white noise auditory stimulus to mask, as much as possible, the sound generated during whisking against the objects. 2) to control for olfactory cues, the same objects were rotated to the same position on the rotary motor, while the apparatus was retracted slightly to be positioned out of reach from the whiskers (termed catch trials).

Our results strongly suggest that mice detected the aluminum foil solely based on the sound that was generated upon whisking. In the task condition (Figure 5D, 'task'), significant differences were found in each pairwise comparison of cues without the additional controls. The significantly higher lick rates (LRs) in the aluminum trials (0.75 ± 0.12 , mean $\pm$ SD, $p < 1e-4$) compared to both the attenuated (0.19 ± 0.07) and no-object (0.11 ± 0.08) trials ($p < 1e-4$)

482 suggests that the sound emitted by the aluminum foil predominantly guided the mice's
483 decisions. In trials with white noise masking the sound, LRs reduced across all cues,
484 particularly blurring the distinction between the attenuated and no-object cues. In the catch
485 trials, where all cues were beyond whisker reach, the LRs were significantly lower and
486 indistinguishable among cues, supporting the hypothesis that auditory cues guided the mice
487 and not other cues. This corroborates our earlier findings (Figure 4) that the three cues are
488 classifiable based on their sound signature. Mice trained to report the presence of aluminum foil
489 detected this object well, although they sometimes mistook the attenuated foil for aluminum foil
490 and licked. The fact that this occasional behavior happens in the attenuated foil trials strongly
491 suggests that mice base their decision on auditory cues. White noise adversely affected the
492 mice's performance in the trials with the two foil types but did not significantly affect the
493 no-object trial LRs.

494 These findings suggest that mice can detect objects through the sound generated during
495 whisking, indicating that this auditory information is functional in task completion. Performance
496 variability under the white noise control condition indicates that while mice can recognize the
497 aluminum object despite the masking noise, their performance is impacted. To appreciate the
498 performance of each mouse in the task, we calculated the d' between cue pairs and across
499 conditions. Overall we find consistent performance of all mice above the expert level ($d' > 1$) in
500 task conditions (Figure 5E), particularly in differentiating between the aluminum and no-object
501 cues ($d' > 1.5$ for all mice), along with the varied performance in the white noise condition
502 (Figure 5F), conclusively demonstrates that mice are capable of detecting and utilizing auditory
503 cues generated during whisking against objects. The low performance of all mice in
504 differentiating between all cues in the catch trials (Figure 5G) suggests that they could not learn
505 the object's presence by other cues (mice were still rewarded if they licked in the aluminum 'go'
506 catch trials). These results affirm the functional relevance of auditory information in guiding
507 behavior, effectively bridging the gap between neuronal responses to sound features and
508 practical behavioral outcomes. This sets the stage for a detailed discussion of the implications
509 of these findings concerning our understanding of auditory processing and behavioral
510 adaptation in mice.



511

512

513 **Figure 5** - Mice detect a specific object based on its whisking-contact sound.

514 (A) Head-fixed mice with their ION cut to prevent tactile sensation whisked against different objects.
 515 Mice were trained to detect aluminum foil by licking a spout to receive a water reward and to avoid
 516 licking when no object was present.

517 (B) Performance of the mice in training sessions. Day 1 marks the introduction of the 'no go' (by
 518 switching the motor to no object position) cue. Before this, the mice were trained exclusively with the
 519 aluminum 'go' cue. For mouse 4, the graph starts from day seven due to technical issues related to data
 520 corruption.

521 (C) Upon achieving performance above the expert level ($d' > 1.5$), the mice were further tested with an
522 additional attenuated foil. Testing included two additional control conditions: masking white noise and
523 catch trials in which objects were retracted out of reach of the whiskers.

524 (D) The average lick rate (LR) for all mice ($n = 5$) is shown, with the plus sign indicating the LR of each
525 mouse. $*p < 0.05$. + indicates individual mice LR.

526 (E) The sensitivity index, d' , between the cues (aluminum-attenuated, aluminum-no object,
527 attenuated-no object) across the testing sessions for each mouse. The line indicates the expert level.

528 (F) Same as (E) for white noise trials.

529 (G) Same as (E) for catch trials.

530

531 Discussion

532 This study reveals that active sensing of objects through the vibrissa system in mice produces
533 sounds that are registered in the auditory system and can be used by the animals during object
534 recognition, challenging conventional sensory modality boundaries. These findings underscore
535 the importance of active behaviors like whisking in sensory integration and decision-making,
536 suggesting a more complex interaction between sensory systems.

537 Traditionally, with the exception of echolocation, the auditory system is viewed as a passive
538 receiver of external sound waves^{32,33}, while tactile sensing demands active exploration^{17,34–36}.
539 Our study demonstrates that sounds generated by whisker movements also aid in object
540 identification, similar to what was found for white cane use¹⁰, suggesting an active component
541 to auditory processing. Our work reveals that the vibrissae system, extensively studied for its
542 tactile functions, is also important for auditory sensing. Our study points to the rodent whisker
543 system as a multimodal sensory processing system that engages both tactile and auditory
544 neural circuits. This finding underscores the sophistication of sensory processing in mammalian
545 nervous systems.

546 We demonstrated that whisking against objects generates sound with unique acoustic
547 signatures for different materials. These sounds fall within spectral ranges and amplitudes
548 audible to mice^{23,24,37–40}. Interestingly, sounds were also generated when whisking without an
549 object. Those sounds varied between animals and were likely produced by body movement.

550 Previous studies have shown that general movement as well as self-initiated sounds can inhibit
551 activity in the AC^{41–46}. Movement can also influence auditory processing at lower levels in the
552 auditory system^{47,48}. Filtering out expected sounds related to body movements offers
553 considerable benefits^{5,6,41,49}. Since the movement of the whiskers is active and controlled by the
554 animal, the auditory system might expect certain acoustic outcomes from it and filter them out.

555 The role of our recording of IC neurons was to validate the hypothesis that the sounds that are
556 generated by whisking against certain objects generate audible sound and we accomplished
557 this goal by completely disconnecting any internal reference for motor signals, and thus to
558 isolate the effect of the sound alone on auditory neurons. Our results confirmed that neurons in
559 the IC, an early auditory processing station, respond to sounds from whisker-object interactions,
560 indicating that these sounds are audible. Using a motor to mimic whisking movements with
561 whiskers plucked from another mouse, we generated sounds which had a similar intensity to
562 that evoked by natural whisking, that elicited neuronal responses in anesthetized mice,
563 confirming these sounds are within the audible spectrum and intensity.

Next, we examined the complex interplay of state, movement, and auditory interactions in awake mice, focusing on neuronal activity in the AC. Auditory processing is affected by other sensory modalities, including the somatosensory system. Some of these interactions occur directly in the cortex, but they can also occur, at lower levels of the ascending auditory pathway^{13,42,49–53}. For example, Lohse et al.⁵⁴ demonstrated a mechanism for somatosensory cortex inhibition mediated by the thalamus. Earlier works by Young et al.¹³ show that tactile stimulation of the cat outer ear (pinna and ear canal) causes direct changes in auditory processing in early auditory processing regions. At multiple stages, there are top-down and bottom-up projections between the somatosensory cortex and the auditory system. On the other hand, little is known about how the AC might affect somatosensation, warranting further research.

Investigating AC neurons' responses to complex whisking-dependent stimuli presents significant challenges. Recent studies on the effects of body movements and projections from motor cortical areas to the AC reveal a complex interplay of excitatory and inhibitory activities across various timescales^{42,51,55–57}. Studies demonstrate that AC activity can decode the running velocity of animals⁴⁴. The considerable influence of running velocity on sound generation and modulation underscores the cortex' role in enhancing the anticipatory functions of the AC.

Our findings reveal that AC activity reflects both the animal's state and the object's identity. Many recorded units exhibited elevated firing rates (FR) during whisking compared to the non-whisking state. Furthermore, self-initiated whisking sounds did not reduce AC activity significantly. Several factors may account for this. First, the aluminum sound was novel to the mice, with no association between the sound and movement expected. Previous research showed that AC response attenuation occurs for specific pure tones, particularly those associated with locomotion over preceding days^{41–43,55,58}. Second, the aluminum produced complex broadband sounds of varying lengths⁵⁹.

To our knowledge, no studies have directly examined the impact of whisking on AC activity. This highlights a crucial aspect of our work: the potential differences in AC responses to artificial sources versus stimuli from direct animal behavior. We suggest that to understand the complex interplay of behavior, activity, sensory and multisensory processing requires designing experiments that can mimic and generate conditions similar to natural ones.

To test behavioral relevance, we trained head-fixed mice on a go/no-go task, where they licked for an aluminum foil cue but refrained from licking for no object. Importantly, the tactile sensation from the whisker pad was eliminated by severing the ION. Mice quickly learned the task, demonstrating they could use auditory cues to guide behavior. White noise masking and catch trials further supported the idea that auditory cues were critical for task performance.

Although our findings reveal a novel aspect of the multimodal sensory process in the rodent whisker system, it is important to acknowledge the limitations of our study. The heterogeneity of our sample, including single isolated units and multi-units, limits our ability to characterize the cellular identity of the AC neurons (such as laminar position, type and projections) that respond to the sounds that were generated by the different conditions. For example, it is well possible that the FR of a specific population of cells was elevated by a given condition while reduced for another population. With regards to the behavior, it is possible that audible sounds are emitted only by certain objects and not others, limiting the ability of mice to explore all objects in their environment by whisking-auditory interactions or when the environment is noisy. Finally, further investigation is necessary to elucidate how whisking generated sounds are processed in ION-intact mice, and how they are integrated with tactile signals processed through the barrel

610 cortex. These observations highlight the need for additional research to refine our
611 understanding of the multimodal nature of the vibrissa system.

612 Although mice are predatory animals and need to minimize exposure, the fact that whiskers
613 generate audible but local and weak sounds can benefit them. If whisking in a certain
614 environment produces a loud sound, moving through it with the entire body will generate
615 significant noise and expose them to predators. Accordingly, mice might use their whiskers to
616 probe environments and decide the best path to minimize exposure risk. Whiskers might also
617 acquire more information on tactilely similar objects. Given that auditory processing responds to
618 higher frequencies than the somatosensory system, a complementary role for both systems in
619 object feature exploration is suggested.

620 Auditory sensing of whiskers may also provide proprioceptive information for whisking behavior.
621 Since whisker muscles and the sensory branch of the whiskers lack muscle spindles or Golgi
622 tendon organs, proprioception of whisking is thought to be predominantly mediated through
623 sensory aspects of contacting air and objects, as well as tactile information from sensory
624 neurons on the skin and fur of the face^{34,35,60,61}. Auditory inputs from whisking can serve as an
625 added form of proprioception to understand whisker location and movements in three
626 dimensions.

627 Our study challenges conventional sensory modality boundaries, advocating for a nuanced
628 approach that embraces the complexity of naturalistic behaviors. Our work lays a foundation for
629 future explorations into sensory systems, highlighting the multimodal and adaptive nature of
630 sensory processing in the mammalian brain.

631 References

632 The following panels and figures were created with [BioRender.com](https://www.biorender.com): Figure 1A; Figure 2A;
633 Figure 3A; Figure 4A; Figure 5A, C; Supplementary Figure 3 – Left;

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827

828 **Methods**

829 Procedures for head-bar surgery and infraorbital nerve severance

830 Unless otherwise stated, all experiments were conducted in female FVB/NHanHsd (Inotiv) mice
831 aged 10-20 weeks. Mice were housed in an enriched cage environment from weaning (aged
832 2-3 weeks), including an Igloo, a fast-track wheel (#K3328,#K3251, Bio-Serv), craft bells and a
833 tunnel made from a Slinky. At 7-8 weeks of age, head-fixation surgery and an infraorbital nerve
834 (ION) cut were performed. Mice were anesthetized using ketamine (140 mg/kg, Zoetis) and
835 xylazine (8 mg/kg, Eurovet). For pain management, buprenorphine (0.1 mg/kg, Richter Pharma)
836 and carprofen (5 mg/kg, Dechra) were injected, and Emla cream (lidocaine/prilocaine, 5%,
837 AstraZeneca) was applied locally. An incision was made along the scalp's midline to expose the
838 skull, the skin was retracted, and the skull was exposed and cleaned. An aluminum head bar,
839 custom-designed and fabricated in our institute's workshop, was affixed to the skull using
840 UV-cured dental cement (Rely-X U200). The head bar was specifically tailored to our
841 experimental requirements. The connecting tissue anterior to the eyes was removed to expose
842 the ION. The entire bundle of the left ION was severed using fine scissors (Vannas-Tübingen
843 #15003-08, FST) as close as possible to the infraorbital foramen, from where it fans out,
844 towards the whiskers. The skin was then reapproximated and sealed using cyanoacrylate
845 adhesive (Vetbond), ensuring the skull was completely covered with no exposed areas.
846 Post-operatively, mice were monitored closely and weighed once a day using a standard
847 laboratory scale for four days, and a recovery period of at least one week was allowed before
848 experiment initiation. Carprofen tablets (2 mg/5 mg, Bio-Serv) were placed in the cage, allowing
849 mice free access for a minimum of three days post-operation.

850 Electrophysiology recordings

851 All electrophysiology recordings were performed with silicone probes (A16 or A32,
852 NeuroNexus) connected to a head stage (RHD2132, Inten). All data from the probe and other
853 sources, camera strobe, microcontrollers, encoder, microphone, and galvanometer movement
854 was collected using the Open Ephys acquisition board (retrieved from
855 <https://open-ephys.org/acquisition-system/acquisition-board>) and the Open Ephys GUI. Data
856 were processed using custom-written Matlab (Mathworks) and Python code. Spike sorting of
857 multi-unit and single-unit was done with kilosort2.5⁶²⁻⁶⁴ or Spike interface⁶⁵ with the following
858 algorithms: Kilosort2.5, KlustaKwik⁶⁶, Ironclust (<https://github.com/flatironinstitute/ironclust>),
859 Wave-clus⁶⁷, Mountain sort (<https://github.com/flatironinstitute/mountainsort>). All spikes were
860 manually curated using Phy2⁶⁸.

861 Sound monitoring and processing

862 The sound was recorded using ultrasonic microphones (FG-Knowles, Avisoft) placed at a
863 distance from the whiskers similar to the mouse's ear distance. The signal from the microphone
864 was processed and amplified using the UltraSoundGate (416H-monitored, Avisoft). It was
865 recorded using the Avisoft USGH software (<https://www.avisoft.com/recorder/>) at 250 kHz
866 sampling frequency (SF), then fed to the Open Ephys acquisition board (30 kHz SF) or a
867 National Instruments DAQ acquisition board (PCIe-6251, PCIe-6343 at 125 kHz SF) for
868 synchronization with other recordings.

869 The microphone's calibration was verified using measurements from a calibrated microphone
870 (4189-W-003, B&K) and a speaker (Vifa, Avisoft) emitting 10 kHz sine wave at 94 dB SPL. The

transformation function for the recorded sound in volts was calculated based on the recorded calibrated signal and the tuning curve of the ultrasonic microphone. In all experiments involving sound recording, an additional analog gain of +30 dB was applied using the UltraSoundGate's predefined gain settings. This specific gain level was selected to optimize data collection capacity while preventing system saturation.

Whisking analysis

The whiskers were filmed from above with a high-speed camera (DMK 37BUX287 camera, The imaging source) at 400 fps and a zoom objective (SMC Pentax 1:1.2/50, Asahi Optical). An IR LED (M940L3, Thorlabs) with a diffuser (DG10-220-MD, Thorlabs) illuminated the whiskers from below for optimal contrast. The whisking video was analyzed using the Facemap GUI^{69,70}. A region of interest in the video (ROI) was chosen manually so that only whiskers were analyzed, and the singular value decomposition (SVD) based on the energy of motion in the ROI was calculated. The whisking data was filtered in the range of 0.1 to 100 Hz (3rd order Butterworth filter) and smoothed with a Gaussian kernel (sigma of 200 ms) to identify whisking epochs. A threshold of 0.7 standard deviations (SDs) above baseline was set to identify whisking epochs, with only continuous data points exceeding this threshold for more than 5 seconds classified as such. Shorter epochs under 5 seconds were excluded to maintain consistency and comparability across events. The selection of 0.7 SDs was based on a thorough analysis of all sound recordings from the study, ensuring optimal detection of whisking events across sessions. This balance was achieved through visual inspection of trace data against baseline noise and validation against high-speed video recordings of whisker movements, confirming the threshold's effectiveness in accurately capturing whisking behavior.

Sounds generated by whisking against objects

Experimental procedure

An awake mouse with a head bar was placed in a custom-built head-fixation apparatus, positioned on a disc treadmill modified from a fast-track wheel (#K3251, Bio-Serv), and connected to a rotary encoder (E6B2-CWZ3E, Yumo). Three objects were utilized to study whisking responses: standard aluminum foil (15 μ m, Diamond), cut into 3 x 1.5 cm rectangles; sound-attenuated aluminum foil, created by adding tape (Magic Tape, Scotch) to standard aluminum foil, maintaining similar tactile feedback but producing less sound; and dried Bougainvillea plant petals, selected for their natural and environmentally relevant stimulus. These objects were placed in a 3D-printed custom-designed holder before the whiskers. For the no object condition, an empty holder was used. The holders were fitted in a custom-designed 3D-printed adapter, facilitating manual switching between objects. The adapter was designed with a circular symmetry to preserve balance and could be occupied by six holders with different objects. Three separate recordings were conducted for each object type, each time using a different instance of the object, ensuring consistency and reproducibility.

Sound processing

Recorded sounds were transformed from voltage into pascals, based on the microphone calibration, and compensated for a +30 dB gain as described previously. Whisking periods were identified as described above. The sound during the whisking periods was separated into epochs based on the condition (Aluminum, attenuated, dry leaf, and no object). Welch's power

spectral density (PSD) was calculated for each epoch of each condition from all recordings and averaged to generate a generalized PSD of the object. Sound epochs from no whisking periods were collected from all recordings to calculate the baseline ambient room noise. The PSD was calculated for all no-whisking periods and averaged across epochs and recordings. Root mean square (RMS) was calculated for each object in the sustained whisking periods. The values were averaged across all whisking epochs from all recordings. Kruskal–Wallis one-way analysis of variance was used with post hoc pairwise comparisons and Tukey-Kramer HSD correction for multi-comparisons.

Sound events identification

The spectrogram was extracted from each whisking epoch for all conditions. A threshold for each epoch was set at the mean plus three standard deviations of the spectrogram. The spectrogram was transformed into a binary mask, with frequency bands above the threshold assigned a value of one, and those below assigned a value of zero (Figure 1C). The temporal start and end points of these events were determined by identifying transitions in the binary mask. Events shorter than 1 ms were removed, and those with less than 80 μ s between them were merged. Only events with at least 30 frequency bands above the threshold were considered for further analysis.

Recordings from the inferior colliculus of anesthetized mice

Experimental procedure

Experiments were conducted using female C57BL/6 mice (Inotiv) aged 6-10 weeks. Urethane (1 g/kg, Sigma) and chlorprothixene (10 mg/kg, Sigma) were administered for anesthesia. Each mouse's skull was exposed and precisely aligned on a stereotactic frame in accordance with the mouse brain atlas⁷¹. A craniotomy of approximately 0.5 mm diameter was performed above the right hemisphere inferior colliculus (IC) at coordinates -5.1 mm anterior-posterior (AP) and +1 mm medial-lateral (ML) relative to Bregma. Whiskers plucked from a deceased mouse, were attached to a glass pipette mounted on a shaft connected to a galvanometer servo control motor (6210H; Cambridge Technology Inc., USA), used in conjunction with a matching servo driver and controller (Micromax 677xx; Cambridge Technology Inc., USA). This setup was positioned approximately 2 cm away from the left ear of the anesthetized mouse. An ultrasonic microphone (FG-Knowles, Avisoft) was positioned right next to the left ear. Various objects, including aluminum foil, attenuated foil, and a no-object control, were presented before the whiskers. Neuronal activity in the IC was recorded using the previously described methodology.

Data analysis and testing

Spiking activity, extracted as previously outlined, was aligned around a window spanning 1 second before to 1 second after the onset of galvanometer movement. The baseline FR, determined from a 110 ms bin one second prior to motor activation, was subtracted from the FR observed in a 110 ms bin immediately following motor onset. This calculation focused on the period encompassing the initial contact with the object, intentionally excluding subsequent contacts to avoid confounding effects of differential adaptation across objects. A Kruskal–Wallis one-way analysis of variance was employed, complemented by Tukey-Kramer HSD correction for multiple comparisons.

For each experimental condition, trial data were segmented into 10 ms bins. This was followed by calculating the recorded responses' peri-stimulus time histogram (PSTH). PSTHs from all units were then averaged, offering a unified view of neural activity across trials. In our statistical analysis, neural activity recorded 110 ms after stimulus onset was adjusted by subtracting the baseline activity for each trial and unit. These adjusted values were averaged across all trials for each unit within each condition. We utilized a Friedman test to determine differences between conditions, again applying Tukey-Kramer HSD correction to accommodate multiple pairwise comparisons.

Recording from the Barrel cortex of infraorbital nerve severed mice

Experimental procedure

A female FVB/NHanHsd mouse, aged eight weeks, was anesthetized using urethane (1 g/kg, Sigma) and chlorprothixene (10 mg/kg, Sigma), and its skull was exposed and aligned on a stereotactic frame in accordance with the mouse brain atlas⁷¹. A precise craniotomy of approximately 0.5 mm diameter was performed above the barrel cortex (BC) in the right hemisphere at coordinates -1.3 mm anterior-posterior (AP) and 3.3 mm medial-lateral (ML) from Bregma. The mouse was placed on a temperature-regulated heating pad, maintaining the body temperature at 36°C. A 32-channel linear silicone probe (A32, NeuroNexus) was inserted through the dural membrane into the barrel cortex to a depth of 1000 µm. A whisker deflection stimulation protocol was initiated once a spiking activity was identified across several channels. Whiskers were deflected using a needle attached to the same galvanometer servo control motor (model as previously mentioned). The stimulation was comprised of three consecutive on-ramp sideways deflections, lasting 25 ms each and at a 10 Hz rate with an 8-second interval between each stimulation. Following sufficient data collection, the probe was gently extracted. Subsequently, the ION was severed as previously described. The probe was then reinserted into the same craniotomy site to a depth of 1000µm, and upon detecting spiking activity, the same stimulation protocol was repeated.

Data analysis and statistical testing

The neuronal data were separated into spiking activity, as previously mentioned, and local field potential (LFP). LFP extraction involved filtering the data between 0.1 and 300 Hz (2nd order Butterworth filter). Motor activity was recorded using the Open Ephys acquisition board to ensure the signal was synchronized with the electrophysiology recordings. Spiking activity from each unit and LFP from each electrode were time-aligned to a window of 0.5 seconds before and after the onset of the whisker deflection.

Spikes from all whisker deflection events were binned in 10-ms time bins to generate a PSTH for each unit. The PSTHs were normalized by dividing the FR by a baseline calculated from the average FR prior to stimulus onset (300 ms window). A right-tailed Wilcoxon rank sum analysis was performed across all recorded units to compare the average activity 100 ms post-stimulus onset before and after the ION severance. The average LFP activity across electrodes, 100 ms post-stimulus onset, were compared employing a right-tailed Wilcoxon sign rank test before and after the ION cut.

Auditory cortex experiments in awake, head-fixed mice

995 Electrophysiology experiments with awake head-fixed mice

996 Animals were prepared in accordance with the previously described methods. Skull alignment
997 was performed in accordance with the mouse brain atlas⁷¹ and marked for recordings from the
998 right hemisphere auditory cortex (AC) at coordinates -2.5 mm anterior-posterior (AP) and 4.5
999 mm medial-lateral (ML) from Bregma prior to affixing the head bar. Cyanoacrylate adhesive
1000 (Vetbond) was then applied to the exposed area of the skull above the recording site. An
1001 aluminum head-bar, custom-made with a 2-mm diameter well, was affixed directly above the
1002 recording site. Over three days preceding the experiment, mice were gradually acclimated to
1003 the experimental setup and head restraint, increasing their daily exposure time to approximately
1004 30 minutes. On the day of the experiment, mice were anesthetized with isoflurane gas (1.5%,
1005 Piamal), and carprofen (5 mg/kg, Dechra) was administered intraperitoneally. An approximately
1006 0.5 mm in diameter craniotomy was performed directly above the recording site. The
1007 craniotomy was sealed with agarose and a silicone elastomer (Body Double, Smooth-On) to
1008 protect the brain. Following a two-hour recovery period post-craniotomy, the mice were
1009 head-fixed and positioned on the disc treadmill as previously described. Subsequently, the
1010 craniotomy was re-exposed, and a silicone probe was inserted perpendicularly through it into
1011 the AC.

1012 Experimental setup

1013 Mice were head-fixed and placed on a disc treadmill as per the previously described methods.
1014 A high-speed (400 fps) infrared camera (DMK 37BUX287, The Imaging Source) was positioned
1015 above the whiskers. Another camera, the same model, operating at 200 fps, was set to capture
1016 the mice's facial and body movements. The cameras' strobe information was streamed to digital
1017 input channels on the Open Ephys acquisition board for synchronization with
1018 electrophysiological data. Initially, a servo motor (HD-1711MG, Power HD) with two arms, with
1019 aluminum and attenuated foil taped onto its arms, was mounted on a linear actuator (LA-T8,
1020 GoMotorWorld.com). Motors were controlled using a microcontroller (Mega 25600, Arduino) to
1021 pseudo-randomly present one of two objects or no object for 60-second periods. Subsequently,
1022 the system was upgraded, and the objects were positioned in a 3D-printed holder, which was
1023 then placed in a 3D-printed adapter with a circular symmetry that could be occupied by six
1024 holders with different objects as previously described. An empty holder space was used for the
1025 no-object condition in one of the six slots. The circular symmetry maintained balance and
1026 allowed a similar presentation of the different objects. This adapter was then mounted on a
1027 stepper motor (1206, Pololu; driver DRV8834, Pololu) and affixed to a linear stepper stage (50
1028 mm, Nema 11; driver A4988, Pololu). These motors were also controlled using a microcontroller
1029 (Teensy 4.0) to pseudo-randomly present one of two objects or no object for 60-second periods.
1030 In the last two sessions, the object presentation duration was reduced to 30 seconds rather
1031 than 60 seconds to optimize data collection. A digital signal indicating the object's identity was
1032 transmitted from the Teensy board to the Open Ephys board's digital inputs. Power for the
1033 camera strobe and motors was supplied by a PCB, designed with EasyEDA⁷² software, and
1034 manufactured by JLCPCB⁷³. Microcontroller and motor drivers were then mounted on and
1035 connected to the PCB. Video recording, experimental control, and data synchronization across
1036 multiple sources were orchestrated using a custom-developed workflow in Bonsai RX⁷⁴.

1037 Data preprocessing

1038 Motion energy was extracted from video footage while spiking activity was derived from the
1039 electrophysiology data as previously described. After starting data acquisition, the camera was
1040 stopped for one second to allow alignment of the camera frames with sound recording and the
1041 electrophysiological data. This gap was identified using the cameras' strobe data collected
1042 through the Open Ephys board's digital inputs. Frame timestamps were gathered using the
1043 custom-developed Bonsai RX workflow. The identified one-second gap in the frame timestamps
1044 was then aligned with the electrophysiology recordings. Sound recording alignment involved a
1045 short burst of white noise emitted from a Vifa speaker (Avisoft), marking both the start and end
1046 of the recording. This white noise, identifiable in the sound recordings from both Avisoft and
1047 Open Ephys, allowed for the synchronization of these recordings. Timestamps were then
1048 interpolated to reconcile differences in sampling rates.

1049 Whisking periods were identified from the extracted motion energy using a threshold as
1050 previously described. Data were subsequently categorized based on whisking activity and the
1051 identity of the objects. Data spanning 3 seconds before and 2 seconds after each object change
1052 were omitted from the analysis to eliminate potential electrical and sound artifacts associated
1053 with motor movement.

1054 Firing rate analysis

1055 After categorizing data by whisking states and objects, spike times were binned into 1 ms bins,
1056 and spike density functions (SDFs) were computed using Gaussian kernels, with a sigma set at
1057 20 ms and edges extending to ± 3 sigma. SDFs were computed for each epoch, each
1058 corresponding to a unique whisking state and object combination. Subsequently, these SDFs
1059 were combined into single vectors to represent each experimental condition. Additionally, SDF
1060 vectors for each object during the no whisking state (NWS) were concatenated into a separate
1061 vector. SDFs covering entire sessions were used to calculate each unit's average firing rate
1062 (FR) and standard deviation (SD), utilizing 1 ms bins. Mean FR and SD, derived from entire
1063 sessions, were used to calculate z-scores for the FR of each condition. Using a Friedman test,
1064 the average FR from all units was then compared across all conditions in the AWS (Active
1065 Whisking State) and NWS separately. Additionally, a one-sided Wilcoxon rank-sum test
1066 compared the average FR differences between the AWS and NWS conditions.

1067 A permutation test comprising 10,000 permutations was conducted for each unit to assess FR
1068 differences across all possible condition combinations. For each pair of conditions, a distribution
1069 of FR differences was generated by permuting the binned FRs of the two conditions 10,000
1070 times. The actual difference between the conditions, expressed in absolute value, was
1071 compared with this distribution of generated differences. The count of generated values equal to
1072 or more extreme than the actual difference was divided by the total number of permutations to
1073 calculate the p-value. The sign of the actual difference indicated the direction of the observed
1074 change. A Tukey-Kramer HSD correction was applied to all calculated p-values for all units and
1075 comparisons, resulting in adjusted p-values. Only differences with an adjusted p-value of less
1076 than 0.05 were considered to be significant.

1077 Classification models

1078 Data preprocessing

1079 FR-based models

SDF data from all units in each session were used to develop classification models aimed at predicting the experimental condition. Average SDFs were computed in 800 ms bins with a 400 ms overlap, and each bin was labeled according to the corresponding experimental condition. For the main model training, data from various objects during AWS were used, while data from all objects during NWS were pooled under a single category. The output labels for the main model included aluminum, attenuated, no object, and no whisking. For the sub-model, labels specific to different objects during NWS were used, resulting in the following labels: NWS aluminum, NWS attenuated, and NWS no object. Model performance, exemplified by a specific session, was analyzed across different time bins (50, 100, 200, 400, 800, and 1000 ms) to identify the optimal bin size. It was observed that performance improved with larger time bins, reaching a peak at 800 ms.

Sound feature models

For the sound-based model, data were categorized using the exact same time bins as the FR-based models. Linear spectra were extracted from each bin at a resolution of 1 kHz. The following sound features^{29–31} were extracted from each bin:

Zero Cross Rate: The rate at which a signal changes from positive to negative or vice versa, indicating frequency content.

Short-Time Energy: The average power of a signal calculated over a short period, indicating the intensity or loudness.

Spectral Centroid: A measure representing the center of mass of the spectrum, perceived as the brightness of a sound.

Spectral Spread: A measure of the spectral width or variance, showing the dispersion of frequencies around the spectral centroid.

Spectral Slope: A measure of the tilt of the sound spectrum, quantifying how spectral energy distribution changes across frequencies.

Spectral Decrease: A descriptor of the rate at which the spectral amplitude decreases with increasing frequency.

Spectral Flux: The rate of change in the power spectrum of a signal, reflecting the speed of changes in the spectral content.

Spectral Crest: A ratio comparing the maximum spectral amplitude to the spectrum's sum, indicating the spectrum's peakiness.

Spectral Flatness: A measure indicating how noise-like a sound is by comparing the geometric mean to the arithmetic mean of the spectral magnitudes.

Spectral Skewness: A measure of the asymmetry of the spectral distribution, indicating if high or low frequencies dominate.

Spectral Kurtosis: A measure of the "tailedness" of the spectral distribution, indicating how outlier frequencies behave.

Spectral Entropy: A measure of the disorder or randomness in the spectral distribution, indicating sound complexity.

Harmonic Ratio: The ratio of harmonic sound components to the total sound, indicating the degree of tonality.

Spectral Roll-off Point: The frequency below which a certain percentage of the total spectral energy is contained, indicating the bandwidth of the signal.

1123 In addition to the specific sound-based model, another model was trained using pooled data
1124 from all experiments. This approach aimed to generalize the model's predictive capabilities
1125 across various experimental setups.

1126 Model training and testing

1127 For each time bin, predictors and labels were split into training (80%) and testing (20%)
1128 datasets, ensuring the label distribution mirrored that of the entire dataset. A consistent
1129 algorithm was employed across all models and sessions to facilitate comparisons. The
1130 performance of various classifier algorithms was evaluated. A model with high performance and
1131 interpretability was preferred to focus on understanding data rather than making predictions on
1132 new data. Based on these criteria, a random forest algorithm was selected^{31,75–77}. Algorithm
1133 hyperparameterization was performed for each session, involving the adjustment of several
1134 parameters to optimize performance:

1135 **Num Learning Cycles:** The number of iterations or rounds for which the ensemble model is
1136 trained affecting the model's ability to learn from the training data.

1137 **Min Leaf Size:** The minimum number of samples required in a leaf node, controlling overfitting
1138 by preventing the tree from growing too deep with overly specific leaves.

1139 **Max Number of Splits:** The maximum number of times a node can be split, determining the
1140 complexity and depth of the tree, influencing both the model's fit and its generalization ability.

1141 Ten-fold cross-validation, capped at 60 objective evaluations, was employed to identify optimal
1142 parameters. The trained model's weights were then applied to predict the labels from the test
1143 dataset. These predictions were then compared with the actual labels to evaluate model
1144 accuracy.

1145 Comparing the agreement of the sound-based and FR-based models

1146 Training and testing data sets were identical between the models (Sound and FR models) to
1147 allow a direct comparison of the sound-based and FR-based models for each session. This
1148 allowed a one-to-one comparison of the prediction for each observation. The predicted labels
1149 for each observation from all sessions were compared, and Choen's kappa agreement score
1150 was calculated using a dedicated Matlab package⁷⁸. A prediction score of each label from all
1151 observations was computed as well. The Pearson correlation coefficient was calculated for
1152 each class based on the prediction scores between the two models.

1153 The same analysis was conducted on a subset of the observations. The subset was computed
1154 by identifying the observations where both models failed to predict the correct label.

1155 Importance extraction

1156 Using a random forest model permits the inclusion of out-of-the-bag (OOB) samples, excluded
1157 from each tree's bootstrap sample in the ensemble. This approach enables assessing model
1158 error by aggregating prediction errors from each tree of the ensemble on its OOB samples. The
1159 OOB allows evaluation of the model performance in a manner similar to cross-fold validation.
1160 By permuting the values of each predictor and then assessing the OOB prediction decrease in
1161 performance, an assessment of the relative importance of each predictor to the model's overall
1162 performance can be extracted. Finally, the importance scores of each predictor were normalized
1163 relative to the maximal importance score of all predictors.

1164 **Go/No go behavioral task**

1165 Experimental set and procedure

1166 Female FVB/NHanHsd (Inotiv) mice, aged 10-20 weeks, were trained on the behavioral task.
1167 Head-bar fixation and infraorbital nerve (ION) cut surgeries were conducted as previously
1168 described. Following at least one week of post-surgery recovery, mice underwent a water
1169 restriction protocol and were given a fixed daily amount of water (0.06 μ l/gram of body weight);
1170 their health and weight were monitored daily for the duration of water restriction. This water
1171 restriction was intended to heighten the mice's motivation to receive a water reward during task
1172 performance. The water deprivation continued until the mice lost at least 10% of their
1173 pre-restriction body weight, which typically occurred after a week. At that point, the mice were
1174 habituated to the experimental set-up previously described by gradually increasing their
1175 head-fixation time in the setup from 10 to 30 minutes over a period of three days. The
1176 experimental control and data collection was done with a custom-designed workflow in Bonsai
1177 RX using two microcontrollers (Teensy 4.0) and a National Instruments DAQ card (PCIe-6251,
1178 PCIe-6343). Mice were monitored using a high-speed IR camera (200 fps, The imaging source)
1179 and video recorded. Objects were presented randomly using the object holders, circular
1180 adapter, stepper motor, and linear stepper stage as previously described. To ensure that the
1181 mouse did not learn the position of the object on the stepper motor and then perform based on
1182 it, the stepper motor turned in both clockwise and anti-clockwise directions randomly before
1183 putting forward the trial object. To further randomize the presentation, each object type in the
1184 holder was duplicated (e.g., two instances of aluminum, attenuated, and no object), and the
1185 software randomly chose which one to present. Before each testing session, the objects and
1186 holder were thoroughly cleaned using distilled water and 70% ethanol to eliminate residual
1187 smells, with the objects being placed in varying positions on the holder. To ensure engagement
1188 in the task, the number of the sucrose water-rewarded trials, namely, the go trials, was the
1189 same as the combined no-go trials (i.e., attenuated object and no object). The running disc was
1190 coated with silicone elastomer (dragon skin, Smooth-on) to reduce noise from the animal
1191 running. The disc angle was adjusted for optimal running conditions and convenience for each
1192 mouse. The experiments were performed in complete darkness with only IR illumination.
1193 After two days of habituation, the mice were introduced to a lick port and rewarded every 1-2
1194 seconds with sucrose water (10% sucrose solution) if they licked the water port. The lick port
1195 was made of a blunt stainless steel needle and connected to a capacitive touch sensor
1196 (SEN-12041, Sparkfun) to detect licks. The quantity of the water reward was controlled
1197 precisely using a pinch solenoid valve (SCH284B003, ASCO). When a mouse licked
1198 consistently for 1-2 training sessions, motor movement, and the aluminum 'go' trials were
1199 introduced. Each trial started with the object positioned outside the whiskers' reach. The object
1200 holder rotated so that one of two aluminum foil objects was chosen randomly and put forward.
1201 Then, the linear stage moved the object holder forward until the aluminum foil made contact
1202 with the whiskers, and the mouse was given 3 seconds to lick; if the mouse licked successfully,
1203 it received a reward of 5-8 μ l of sucrose water (10% sucrose solution). To keep the mouse
1204 motivated and help it learn the task, a variable percentage of trials were free water trials (based
1205 on animal performance) where the mouse received a water reward even if it did not lick. Once
1206 the mouse reached a satisfactory performance level, i.e., licked over 80% during aluminum 'go'
1207 trials, indicating task comprehension, no-object 'No-go' trials were gradually introduced. To
1208 motivate the mice to lick only during appropriate trials, the trials were spaced with a long (>15
1209 sec) inter-trial interval (ITI). Broadband white noise was emitted from an ultrasonic speaker

1210 (Vifa, Avisoft) in a small percentage of the trials (3%). The presentation of the white noise
1211 during the training stage allowed the animals to habituate and limit startling in response to it. As
1212 a result, when the percentage of white noise trials increased during the testing stage, it was
1213 more likely that any observed effect was due to the masking of whisker-touch sounds rather
1214 than a startle response.

1215 Once a mouse reached a high-performance level ($d' > 1.5$) for three consecutive days, its
1216 performance was tested. At the testing stage, an attenuated foil 'no-go' trial was added, along
1217 with two control conditions: white noise and catch. The control trials aimed to test whether the
1218 mice used cues other than the whisker-touch sounds to identify the objects. The white noise
1219 served to mask the whisker-touch sounds, while during the catch trials, the linear stage stopped
1220 before the object made contact with the whiskers. In each testing session, 30% of trials were
1221 control trials, either white noise or catch. Each mouse was tested 3 to 4 times to collect enough
1222 data for all objects in the task and in control trials.

1223 Lick rate comparisons permutation testing

1224 The number of trials that the mouse licked for each cue in each condition was calculated to
1225 assess the overall performance across animals. A Repeated Measures ANOVA test was
1226 performed, and the F-statistics values for the interaction effect and the main effects of cues and
1227 conditions were extracted. A permutation test was designed to compute the likelihood of
1228 receiving an F-statistics equal to or larger than the calculated F-statistics of the interaction and
1229 main effects. The following steps were performed to compute the p-value for the interaction
1230 effect:

- 1231 • For each mouse, the interaction term was computed for each cue and condition
1232 combination based on the lick rate(LR):
1233
$$Interaction_{cue,condition} = LR_{cue,condition} - (Mean_LR_{cue} + Mean_LR_{condition} - GrandMean_LR)$$

1234 • The interaction term was permuted within an animal to maintain the within animal
1235 structure.
- 1236 • The permuted interaction terms were used to construct a permuted data set:
1237
$$Permuted_LR_{cue,condition} = Mean_LR_{cue} + Mean_LR_{condition} + PermutedInteraction_{cue,condition} -$$

1238
$$GrandMean_LR$$
- 1239 • A Repeated Measures ANOVA was performed, and an F-statistics for the permuted
1240 interaction effect was extracted.
- 1241 • The procedure was repeated for 10,000 permutations.
- 1242 • The observed F-statistic was compared with a permuted F-statistic, and the number
1243 of values that are as or more extreme compared to the observed F - F-statistic was
1244 counted and divided by the total number of permutations to receive a p-value of the
1245 interaction effect.

1246 To compute the cue and condition main effects p-values. A permutation distribution (10,000
1247 permutations) was created by shuffling either the cue labels within each condition or the
1248 condition labels within each cue, respectively, within each mouse. A Repeated measures
1249 ANOVA was performed, and the F-statistics for each main effect were extracted from the
1250 relevant permutation. The observed F-statistics for the relevant main effect were compared with
1251 the permutations distribution, and the number of times the values were as or more extreme than
1252 the observed value were counted and divided by the total number of permutations to compute
1253 the p-value.

1254 Next, based on the results of the repeated measures ANOVA, the relevant post hoc analysis for
1255 simple effects was performed. A one-way ANOVA test was performed to calculate the

1256 F-statistics of the simple effects for the cue within each condition and the simple effects for the
1257 condition within each cue. The permutation distributions obtained from the main effect were
1258 used to calculate the F-statistics of the simple effects and compare them with the calculated
1259 simple effects F-statistics. Values as or more extreme than the actual value were counted, and
1260 the sum was divided by the total number of permutations to compute a p-value. The p-values
1261 were adjusted for multiple comparisons using the Bonferroni method.

1262 The relevant pairwise comparisons based on the results of the simple effects tests were
1263 performed. Again, using the same permutation distributions calculated for the main effect, the
1264 actual LR difference between each relevant pair-wise comparison was compared with the LR
1265 differences calculated from the permuted distribution. Permuted LR differences that were as or
1266 more extreme compared to the actual LR difference were counted, and the sum was divided by
1267 the number of permutations to receive a p-value. The p-values were adjusted for multiple
1268 comparisons using the Bonferroni method.

1269 Calculating d'

1270 The outcome (whether a mouse licked or not) was recorded for each trial. Licking in go trials
1271 (aluminum) was classified as a hit, and in no-go trials (attenuated or no object), it was classified
1272 as a false alarm. d' was calculated for each pair of cues within each condition across mice. Hit
1273 Rate (HR) was determined by the ratio of Hits to the sum of Hits and Misses, while False Alarm
1274 (FA) Rate was calculated as the ratio of FAs to the sum of FAs and Correct Rejections.
1275 Aluminum cue trials consistently served as the target (Hits) in the comparisons. For
1276 comparisons between attenuated and no-object cues, attenuated trials were designated as the
1277 target (Hits). d' was computed by subtracting the z-score of the FA rate from the z-score of the
1278 HR.